

EXAM 7 – SPRING 2013

1. (3 points)

Given the following information:

Age-to-Age Loss Development Factors

	<u>12-24</u>	<u>24-36</u>	<u>36-48</u>	<u>48-60</u>	<u>60-72</u>
2007	2.45	1.73	1.19	1.05	1.05
2008	5.42	1.26	1.23	1.09	
2009	2.64	1.35	1.16		
2010	2.04	1.55			
2011	6.23				

z-value for 90th percentile of the normal distribution: 1.645

a. (2.5 points)

The null hypothesis is that the triangle does not display calendar year effects. Conduct a test to determine whether the null hypothesis should be accepted or rejected at the 90% confidence level.

b. (0.5 point)

Briefly describe two potential causes of calendar year effects in loss development data.

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Question 1 Sample Answer

Solution 1

a)

	<u>12-24</u>	<u>24-36</u>	<u>36-48</u>	<u>48-60</u>	<u>60-72</u>
2007	S	L	*	S	*
2008	L	S	L	L	
2009	*	S	S		
2010	S	L			
2011	L				

S= Less than median L=Greater than median*= median

<u>Diagonal</u>	<u>#S</u>	<u>#L</u>	<u>Z</u>	<u>n</u>	<u>m</u>	<u>c_n</u>	<u>E[Z_n]</u>	<u>VAR[Z_n]</u>
2	0	2	0	2	0	.5	.5	.25
4	3	1	1	4	1	.75	1.25	.4375
5	1	3	<u>1</u>	5	1	.75	<u>1.25</u>	<u>.4375</u>
Total			2				3	1.125

$$Z = \min(\#S, \#L) \quad n = \# \text{ of elements} \quad m = \text{floor}\left[\frac{n-1}{2}\right] \quad c_n = \binom{n-1}{m} \frac{n}{2^3}$$

$$E[Z_n] = \frac{n}{2} - c_n \quad \text{var}[Z_n] = \frac{n(n-1)}{4} - (n-1)c_n + E[Z_n] - (E[Z_n])^2$$

Test statistic =2

$$90\% \text{ CI} = 3 \pm 1.645(\sqrt{1.125}) = (1.26, 4.74)$$

Since z is within range, accept H_0 and conclude there is not a calendar year effect

b)

* A Change on claim handling could cause this. For instance, the claims department might decide to strengthen outstanding case reserves for all claims in a given calendar year

* Legislative Changes: A law change might impact all claims settled after that date

Solution 2

a)

Development Period	<u>12-24</u>	<u>24-36</u>	<u>36-48</u>	<u>48-60</u>	<u>60-72</u>
Median Age to Age Factor	2.64	1.45	1.19	1.07	1.05

	<u>12-24</u>	<u>24-36</u>	<u>36-48</u>	<u>48-60</u>	<u>60-72</u>
2007	L	H	-	L	-
2008	H	L	H	H	
2009	-	L	L		
2010	L	H			
2011	H				

L= Less than median H=above the median*= median

CY Diag	<u>#H</u>	<u>#L</u>	<u>Z</u>	<u>n</u>	<u>m</u>	<u>c_n</u>	<u>E[Z_n]</u>	<u>VAR[Z_n]</u>
2007	0	1						
2008	2	0	0	2	0	.50	0.50	.250
2009	0	1						
2010	1	3	1	4	1	.75	1.25	.4375
2011	3	1	<u>1</u>	4	1	.75	<u>1.25</u>	<u>.4375</u>
Total			2				3	1.125

N= # of L's and H's

$$Z = \min(\#L, \#L) \quad n = \# \text{ of elements} \quad m = \text{floor}\left[\frac{n-1}{2}\right] \quad c_n = \binom{n-1}{m} \frac{n}{2^3}$$

$$E[Z_n] = \frac{n}{2} - c_n \quad \text{var}[Z_n] = \frac{n(n-1)}{4} - (n-1)c_n + E[Z_n] - (E[Z_n])^2$$

$$CI = E(z_n) \pm z_{90\%} \sqrt{\text{Var}(z)}$$

$$CI = 3 \pm 1.645 \sqrt{1.125} = (1.2552, 4.74487)$$

Our z=2 is within the 90% confidence interval so we accept the null hypothesis that the triangle does not display calendar year effects

b)

* A change in the legal environment could affect all the claims in a CY

* If the claims department strengthened reserves for all open claims that would increase the ldf's for that CY.

Examiner Comment

Part a) Most candidates were able to correctly create the S/L/* table, but failed to mention the median or incorrectly used the mean. Some candidates used the rankings based on the calendar years and not accident years.

Candidates were expected to calculate the z , $E(z)$, and $\text{Var}(z)$ as well as provide formulas for calendar years with at least 2 elements. Common mistakes included using the wrong diagonal, not calculating the mean and variance correctly, and not providing formulas. Credit was granted as long as the numbers were consistent with the S/L/* table.

Most candidates were able to provide the confidence interval and provide accept/reject decision correctly. Common errors included missing the square root, not using the correct test statistic, and incorrect decision. Credit was granted if answers were consistent with previous steps.

Credit was given whether the CI was calculated based on the z -value of 1.645 provided in the question or 1.96.

We provided limited credit for candidates that described how to calculate the CI and the accept/reject criterion, but did not provide numbers.

b) We accepted standard comments on changes in inflation/trends, claim reserving practices, and legislative/judicial changes that would have a calendar year impact.

We also accepted answers with adequate explanation on how they would impact calendar year results.

The common mistakes were the same type of item twice and also including items that would cause accident year effects and not calendar year.